Car Sales Dashboard – Power BI Case Study

GitHub Repository:

https://github.com/direct2subhajit/Power_BI_Projects/tree/main/Car_Sales_Analysis

Project Overview

This project focuses on building an executive-level Car Sales Dashboard using Power BI to deliver real-time insights into revenue performance, sales volume, pricing trends, and regional contribution. The objective was to transform fragmented reporting into a centralized, interactive Business Intelligence solution aligned with modern analytics standards.

Business Challenge

The dealership lacked structured visibility into Year-to-Date (YTD), Month-to-Date (MTD), and Year-over-Year (YoY) metrics. Leadership teams could not efficiently compare performance across regions, product categories, or time periods, limiting proactive decision-making.

Technical Approach

- Designed a scalable Star Schema data model (Fact Sales with supporting dimension tables).
- Created a dedicated Date table to enable robust time-intelligence analysis.
- Developed advanced DAX measures including TOTALYTD(), TOTALMTD(), SAMEPERIODLASTYEAR(), and YoY Growth calculations.
- Built dynamic KPI cards with conditional formatting to highlight performance trends.
- Designed interactive visuals: Weekly Sales Trend, Body Style & Color Distribution, Regional Map, and Company Performance Grid.

Key Metrics & Results

- YTD Total Sales: $371.2M with 23.59% YoY growth (+$70.8M).
- YTD Cars Sold: 13.3K units (+19.73% YoY).
- YTD Average Price: $28.0K (-0.79% YoY).
- Monthly Performance: $54.28M sales and 1.92K cars sold (MTD).
- Weekly Revenue Peak: ~$14.9M.
- SUV identified as highest revenue-generating segment.
- Pale White emerged as the most preferred vehicle color.

Business Impact

- Enabled real-time KPI monitoring for leadership.
- Improved visibility into volume-driven revenue growth.
- Supported pricing analysis and competitive positioning.
- Highlighted high-performing regions and product segments for strategic focus.
- Demonstrated readiness to deliver scalable BI solutions in fast-paced analytics environments.